A2 Dossier Snapshot: Spending Analyzer

Team: Josh Ramirez, Neil Paul, Neil Shrestha · CSCI 4150, Fall 2026

We're building a Spending Analyzer for college students. Each entry below changed what we built next. In A2 we tested those ideas side by side.

Entry	What we tested	What we found	What it changed
Meeting 5 (Sep 15)	Text classifier on urgent messages	Called "I may need an extension because of illness" not urgent	Word-based models miss context
Sep 22	Picked the Spending Analyzer; similarity search test	Two different questions got the same result and the same score (0.6191)	A similarity score shows text is related, not that it's right
Meeting 11	Would a neural network help?	Messy codes, rare charges, receipts and too little data	MODIFY; look into OCR for receipts
Oct 9	LLM reading spending and making a budget	Good with the data, guessed without it; might make math mistakes	MODIFY: code does the totals, the LLM categorizes and explains
A2	Every approach on the same 68 September charges	Trained models fell from about 85% to 25% on new stores; LLM got 93–94% and flagged 17 of 19 misses	Adopt code for the math and the LLM to sort charges, asking the student when it's unsure

What changed our mind: on new stores, the classifier's most confident guesses (above 0.70) were right 0 of 28 times. We deferred the neural net and OCR. Our next question is whether a student's past answers can settle mixed stores like Target.

Where to find the rest: the system map is slide 3, the evidence table is slide 10, and the raw data, notebook and LLM replies are in the "A2 evidence files" folder.